

Frederic Rosen, Algebra of Mohammed Ben Musa (1831)

Islamic Original

Modern LaTeX Editions of Public-Domain Mathematics Manuscripts

الحساب علم مبادي شرح

الظنون كشف

الابي ابن

والمقابلة الجبر في مسلم بيد صنفه كتاب اول هذا

فصحه بغداد. الى الروم بلاد من راجعا موسى بن محمد عليه قدم ان الى بها فاقام كفرطابا وسكن حران من خرج قرة بن ثابت
الحجاب اهل جملة من وكان المتوكل الى وقربه داره فنزله بغداد الى فحملة وذكاءه فضله وعرف معرفته في واجتهد
المزيحفي

$$1^{st}cx^2 + bx = a$$

$$2^d cx^2 + a = bx$$

$$3^d cx^2 = bx + a$$

$$1^{st}cx^2 + bx = a$$

$$x^2 + 10x = 39$$

$$x = \sqrt{25 + 39} - 5 = \sqrt{64} - 5 = 8 - 5 = 3$$

$$cx^2 + bx = ax^2 + \frac{b}{c}x = \frac{a}{c}$$

$$2x^2 + 10x = 48x^2 + 5x = 24$$

$$x = \sqrt{6\frac{1}{4} + 24} - 2\frac{1}{2} = \sqrt{30\frac{1}{4}} - 2\frac{1}{2} = 5\frac{1}{2} - 2\frac{1}{2} = 3$$

$$\frac{1}{2}x^2 + 10x = 56x^2 + 10x = 56x = \sqrt{25 + 56} - 5 = \sqrt{81} - 5 = 9 - 5 = 4$$

$$2^d cx^2 + a = bx$$

$$x^2 + 21 = 10x$$

$$x = \frac{10}{2} \pm \sqrt{\left(\frac{10}{2}\right)^2 - 21} = 5 \pm \sqrt{25 - 21} = 5 \pm \sqrt{4} = 5 \pm 2 = 73$$

$$x^2 + a = bx\left(\frac{b}{2}\right)^2 < a\left(\frac{b}{2}\right)^2 = ax = \frac{b}{2}$$

$$\left(\frac{b}{2}\right)^2 < a$$

$$cx^2 + a = bxx^2 + \frac{a}{c} = \frac{b}{c}x$$

$$3^d cx^2 = bx + a$$

$$x^2 = 3x + 4$$

$$x = \sqrt{\left(\frac{3}{2}\right)^2 + 4} + \frac{3}{2} = \sqrt{2\frac{1}{4} + 4} + 1\frac{1}{2} = \sqrt{6\frac{1}{4}} + 1\frac{1}{2} = 2\frac{1}{2} + 1\frac{1}{2} = 4$$

ABABCGTKABDHDHAB

	<i>G</i>	
<i>C</i>	<i>AB</i>	<i>T</i>
	<i>K</i>	

ABABGDABABSHAB

<i>D</i>	
<i>AB</i>	<i>G</i>

ADADHNHBHCHCCDCHGGCHGGTCDGTCGGTTKKMMTTK
HBHBKTMTTAKMKLGKTGMLKCLKMKGMRTAHTMRHB
MTMTHTMKRKRRGGACGACCGCR

AD
HDADHCRDHBCHGHTGTAHTLGLAGKNTLGMAGMLAGGLCGNRMNTLHBKL

ARANKLAR
GMANKL
ADGCAGGMCGACAD

$$x^2 + 10x = 39$$
$$4 \times \left(\frac{b}{4}\right)^2 = \left(\frac{b}{2}\right)^2$$

$$\begin{aligned}
&xyxy \\
&x \pm ay \pm bxyxbayab \\
&y \pm x = a \pm b \\
(+a)(+b) &= +ab(-a)(-b) = +ab(+a)(-b) = -ab(-a)(+b) = -ab \\
(10+1) \times (10+2) &= 10 \times 10 + 1 \times 10 + 2 \times 10 + 1 \times 2 = 100 + 10 + 20 + 2 = 132 \\
(10-1) \times (10-1) &= 10 \times 10 - 10 - 10 + 1 = 100 - 20 + 1 = 81 \\
(10+2) \times (10-1) &= 10 \times 10 - 10 + 20 - 2 = 100 - 10 + 20 - 2 = 108 \\
(10-x) \times 10 &= 10 \times 10 - 10x = 100 - 10x \\
(10+x) \times 10 &= 10 \times 10 + 10x = 100 + 10x \\
(10+x)(10+x) &= 10 \times 10 + 10x + 10x + x^2 = 100 + 20x + x^2 \\
(10-x)(10+x) &= 10 \times 10 - 10x + 10x - x^2 = 100 - x^2
\end{aligned}$$

$$\begin{aligned}
&\sqrt{200} - 10 + (20 - \sqrt{200}) = 10 \\
&20 - \sqrt{200} - (\sqrt{200} - 10) = 30 - 2\sqrt{200} = 30 - \sqrt{800} \\
&50 + 10x - 2x^2 + (100 + x^2 - 20x) = 150 - 10x - x^2 \\
&100 + x^2 - 20x - [50 - 2x^2 + 10x] = 50 + 3x^2 - 30x \\
&(2)^2 \times x^2 = 4x^2 \sqrt{4x^2} = 2x \\
&3\sqrt{9} = \sqrt{9 \times 9} = \sqrt{81} = 9 \\
&3\sqrt{9} = \sqrt{9 \times 9} = \sqrt{81} = 9 \\
&\frac{1}{2}\sqrt{9} = \sqrt{\frac{1}{4} \times 9} = \sqrt{2\frac{1}{4}} = 1\frac{1}{2}
\end{aligned}$$

$$\begin{aligned}\frac{\sqrt{9}}{\sqrt{4}} &= \sqrt{\frac{9}{4}} = \frac{3}{2} = 1\frac{1}{2} \\ \frac{\sqrt{4}}{\sqrt{9}} &= \sqrt{\frac{4}{9}} = \frac{2}{3} \\ \frac{2\sqrt{9}}{\sqrt{4}} &= \frac{\sqrt{4} \times \sqrt{9}}{\sqrt{4}} = \sqrt{9} = 3 \\ \sqrt{4} \times \sqrt{9} &= \sqrt{4 \times 9} = \sqrt{36} = 6 \\ \sqrt{10} \times \sqrt{5} &= \sqrt{5 \times 10} = \sqrt{50} \\ \sqrt{\frac{1}{3}} \times \sqrt{\frac{1}{2}} &= \sqrt{\frac{1}{3} \times \frac{1}{2}} = \sqrt{\frac{1}{6}} \\ 3\sqrt{4} \times 2\sqrt{9} &= \sqrt{9 \times 4} \times \sqrt{4 \times 9} = \sqrt{36 \times 36} = 36\end{aligned}$$

ABACABCBBDACBHABHD
CBHDBHCBSHABBHACSBABCBBHSHSHHDBDACBSSD

ABACBDBHABCB
CBHDBSACSDSBBB
HDCBHGGDSDBHSBBCHGCBSD

$$(10-x)x = x^2 = 4(10-x)x = 40x - 5x^2 \quad 5x^2 = 40xx^2 = 8xx = 8(10-x) = 2$$

$$100 = x^2 \times 2\frac{7}{9} \frac{1}{2\frac{7}{9}} \times 100 = x^2x = 610 - x = 4$$

$$2\frac{7}{9} = \frac{25}{9} \frac{1}{\frac{25}{9}} \times 100 = \frac{9}{25} \times 100 = 36x^2 = 36x = 6$$

$$\frac{10-x}{x} = 410 = 5xx = 2(10-x) = 8$$

$$(\frac{1}{3}x + 1)(\frac{1}{4}x + 1) = 20\frac{1}{12}x^2 + \frac{7}{12}x + 1 = 20x^2 + 7x = 228$$

$$(10-x)^2 + x^2 = 58100 - 20x + x^2 + x^2 = 582x^2 - 20x + 100 = 58x^2 - 10x + 50 = 29x = 5 \pm \sqrt{25 - 29} = 5 \pm 2 = 7$$

3

$$\frac{1}{3}x \times \frac{1}{4}x = x + 24\frac{1}{12}x^2 = x + 24x^2 = 12x + 288x = 6 + \sqrt{36 + 288} = 6 + 18 = 24$$

$$\begin{aligned}
(10-x)x &= 2110x - x^2 = 21x = 5 \pm \sqrt{25-21} = 5 \pm 2 \\
(10-x)^2 - x^2 &= 40100 - 20x + x^2 - x^2 = 40100 - 20x = 40100 = 20x + 4060 = 20xx = 3 \\
100 - 20x + x^2 + x^2 + 10 - 2x &= 54100 - 22x + 2x^2 = 5455 - 11x + x^2 = 27x^2 + 28 = 11xx = \frac{11}{2} \pm \sqrt{\frac{121}{4} - 28} = \\
\frac{11}{2} \pm \frac{3}{2} &= 74 \\
100 + 2x^2 - 20x &= x(10-x) \times 2\frac{1}{6} = 21\frac{2}{3}x - 2\frac{1}{6}x^2 100 + 4\frac{1}{6}x^2 = 41\frac{2}{3}x24 + x^2 = 10xx = 5 - \sqrt{25-24} = 5-1 = 4 \\
6 \quad 4\frac{1}{6} &= \frac{25}{6} \frac{1}{4\frac{1}{6}} = \frac{6}{25} = \frac{1}{5} + \frac{1}{5} \frac{1}{5} \\
\frac{a}{b} \times \frac{b}{a} &= 1 \\
x^2 + 100 &= 50xx = 25 - \sqrt{625-100} = 25 - \sqrt{525} \\
100 - 20x + x^2 &= 81xx^2 + 100 = 101xx = \frac{101}{2} - \sqrt{\frac{10201}{4} - 100} = 50\frac{1}{2} - 49\frac{1}{2} = 1 \\
mnrxxmrx + nx &= (m-n) + (rx-x)x = \frac{m-n}{mr+n-r+1} \\
\frac{x}{x+2} &= \frac{1}{2}(x+2) = x\frac{1}{2}x + 1 = x1 = \frac{1}{2}xx = 24 \\
10x &= (10-x)^2 = 100 - 20x + x^2 \\
\frac{(10-x)x}{10-2x} &= 5\frac{1}{4}20\frac{1}{4}x = x^2 + 5\frac{1}{4}(10-2x)x = 10 - 7\frac{1}{2} = 2\frac{1}{2} \\
\frac{2}{3} \cdot \frac{1}{5}x^2 &= \frac{1}{7}x\frac{2}{15}x^2 = \frac{1}{7}xx^2 = \frac{15}{14}xx = \frac{15}{14} \\
\frac{3}{4} \cdot \frac{1}{5}x^2 &= \frac{4}{5}x\frac{3}{20}x^2 = \frac{4}{5}x\frac{3}{20}x = \frac{16}{5}x = \frac{16}{3} = 5\frac{1}{3} \\
4x^2 &= 20x^2 = 5x = \sqrt{5} \\
x \times \frac{x}{3} &= 10x^2 = 30x = \sqrt{30} \\
x \times 4x &= \frac{1}{3}x4x^2 = \frac{1}{3}x12x^2 = xx = \frac{1}{12} \\
x^2 \times x &= 3x^2x^3 = 3x^2x = 3 \\
4x \times 3x &= x^2 + 4412x^2 = x^2 + 4411x^2 = 44x^2 = 4x = 2 \\
4x \times 5x &= 2x^2 + 3620x^2 = 2x^2 + 3618x^2 = 36x^2 = 2 \\
x \times 4x &= 3x^2 + 504x^2 = 3x^2 + 50x^2 = 50x = \sqrt{50}
\end{aligned}$$

$$x^2 + 20 = 12xx = 6 \pm \sqrt{36 - 20} = 6 \pm 4 = 102$$

$$(x^2 - \frac{1}{3}x^2 - 3)^2 = x^2 \frac{2}{3}x^2 - 3 = x \frac{4}{9}x^4 - 4x^2 + 9 = x^2x^2 + 9 = 5xx = 92\frac{1}{4}$$

$$\frac{1}{3}x \times \frac{1}{4}x = x \frac{1}{12}x^2 = xx^2 = 12xx = 12$$

$$\frac{x}{3} + 1 \times \frac{x}{4} + 2 = x + 13\frac{1}{12}x^2 + \frac{11}{12}x + 2 = x + 13\frac{1}{12}x^2 + 11 = \frac{1}{12}x + 13x^2 = 132 + xx = 12$$

$$xx^2 + x = \frac{3}{2}x = \sqrt{1\frac{1}{4} + \frac{1}{2} - \frac{1}{2}}$$

$$(x - \frac{1}{3}x - \frac{1}{4}x - 4)^2 = x + 12\frac{5}{12}x - 4x^2 + 23\frac{1}{4} = 24\frac{1}{4}x$$

$$x \times \frac{2}{3}x = 5\frac{2}{3}x^2 = 5x^2 = 7\frac{1}{2}x = \sqrt{7\frac{1}{2}}$$

$$\frac{x}{x+2} = \frac{1}{2}\frac{1}{2}(x+2) = x1 + \frac{1}{2}x = x1 = \frac{1}{2}xx = 24$$

$$\frac{1}{x} - \frac{1}{x+1} = \frac{1}{6}\frac{x+1-x}{x(x+1)} = \frac{1}{6}\frac{1}{x(x+1)} = \frac{1}{6}x^2 + x = 6x = \sqrt{\frac{1}{4} + 6} - \frac{1}{2} = 2$$

$$\frac{2}{3}x^2 = 5x^2 = 7\frac{1}{2}x = \sqrt{7\frac{1}{2}}$$

$$x^2 \times 3x = 5x^23x^3 = 5x^23x = 5x = 1\frac{2}{3}x^2 = 2\frac{7}{9}$$

$$(x^2 - \frac{1}{3}x^2) \times 3x = x^2\frac{2}{3}x^2 \times 3x = x^22x^3 = x^2x = \frac{1}{2}x^2 = \frac{1}{4}$$

$$x^2 - 4x = \frac{1}{3}(x^2 - 4x)\frac{2}{3}(x^2 - 4x) = 4xx^2 - 4x = 6xx^2 = 10xx = 10x^2 = 100$$

$$\sqrt{x^2 - x} + x = 2\sqrt{x^2 - x} = 2 - xx^2 - x = 4 + x^2 - 4x3x = 4x = 1\frac{1}{3}x^2 = 1\frac{7}{9}$$

$$x^2 - 3x = xx^2 = 4xx = 4$$

$$abABa : b :: A : BaB = bA \cdot a = \frac{bA}{B} b = \frac{aB}{A}$$

ABCDACHRABTGABCDKCKBKDKHTAKATHHKTATABAHACTHTRRGGH
ADATAKHTATAHTH

$$\begin{aligned} 1^{st} \frac{3}{4} d &= p3.1428 d 2^d \sqrt{10 d^2} = p3.16227 d^{\frac{d \times 62832}{20000}} = p3.1416 d \\ d\pi \frac{d^2}{4} &= d^2 - \frac{d^2}{7} - \frac{1}{2} \cdot \frac{d^2}{7} = d^2(1 - \frac{3}{14}) = \frac{11}{14} d^2 \approx 0.7857 d^2 \frac{22}{7} \cdot \frac{1}{4} = \frac{11}{14} \approx 0.7857 \\ dd' s \frac{dd'}{2} &= d \times \sqrt{s^2 - \frac{d^2}{4}} \\ \sqrt{10^2 - 5^2} &= \sqrt{75} \sqrt{75} = 25 \sqrt{543.3} + \\ \sqrt{169 - x^2} &= \sqrt{29 + 28x - x^2} 169 = 29 + 28x 140 = 28xx = 5 \end{aligned}$$

$$\begin{aligned}
& n n x^{\frac{2}{3}}[10+x] = 2x. \therefore x = 5 \\
& \frac{4}{5}[10+x] - 1 = 2x. \therefore \frac{4}{5}[10+x] - 1 = 2x^{\frac{2}{3}}[10+x] - 1 = x^{\frac{1}{5}}[10+x] + 1 \\
& \frac{1}{5}[10+x] - 1 \\
& \frac{1}{5} \cdot \frac{1}{6} + \frac{1}{6} = \frac{7}{24} \cdot \frac{17}{24} \cdot \frac{17}{48} = 1 = v 1 - \frac{1}{9} = \frac{8}{9} \cdot \frac{8}{9} = 48v. \therefore v = \frac{1}{54} = \frac{1}{54} \\
& \frac{1}{3} \cdot \frac{1}{3} \cdot \frac{1}{3} = \frac{3}{20} \cdot \frac{6}{20} \cdot \frac{1}{8} + \frac{1}{7} = \frac{15}{56} = \frac{41}{56} \cdot \frac{1}{20} = \frac{41}{56} \times \frac{1}{20} = \frac{41}{1120} \cdot \frac{15}{56} = \frac{300}{1120} = \frac{15}{56} \\
& \frac{1}{4} \cdot \frac{1}{6} + \frac{1}{6} = \frac{5}{12} \cdot \frac{7}{12} \cdot \frac{2}{5} + \frac{1}{4} = \frac{13}{20} \\
& \frac{1}{8} \cdot \frac{7}{8} x = \frac{1}{8}(1-x) = \frac{1}{4} \cdot \frac{7}{8}(1-x) = \frac{7}{32} - \frac{4}{32}(1-x) = x. \therefore x = \frac{3}{35} \\
& \frac{2}{7} x^{\frac{2}{3}}(1-x) = x. \therefore x = \frac{2}{9} 1 - x = \frac{2}{9} = \frac{2}{7} \cdot \frac{7}{9} = \frac{2}{9} = \frac{1}{7} \cdot \frac{7}{9} = \frac{1}{9} = \frac{2}{9} \\
& x^{\frac{1}{8}} \cdot \frac{7}{8}(1-x)^{\frac{1}{8}} [1-x]^{\frac{1}{7}} \cdot \frac{7}{8}(1-x)^{\frac{1}{8}} \cdot \frac{8}{8}(1-x) = \frac{1}{7} \cdot \frac{7}{8}(1-x) - \frac{1}{8}(1-x) = x. \therefore x = \frac{45}{336} \\
& \frac{2}{13} \cdot \frac{1}{13} 1 - x - y = 13v 1 - [\frac{1}{5} - v] - [\frac{1}{4} - 2v] = 13v 1 - \frac{1}{5} + v - \frac{1}{4} + 2v = 13v \\
& s d x d + x^{\frac{1}{3}}[d+x]^{\frac{2}{3}}[d+x]^{\frac{1}{3}}[d+x]. \therefore s - [d+x] + \frac{1}{3}[d+x] x s - \frac{2}{3}[d+x] = 2x. \therefore \frac{3}{8}[3s-2d] = x s = 100. \therefore x = 35 \\
& d+x = 45\frac{1}{3}[d+x] = 152x = 70
\end{aligned}$$

$$\begin{aligned}
& pu\frac{2}{3}[u-p]\frac{1}{3}[u-p]\frac{1}{2}[u-p]\frac{2}{3}p\frac{1}{3}p\frac{1}{2}[u-p]\frac{1}{2}[u-p] \\
& \frac{2}{3}\times\frac{1}{3}=\frac{2}{9} \\
& auxu+x-a\frac{1}{3}[u+x-a]\frac{1}{2}[u+x-a]a-x\frac{1}{3}[u+x-a]\frac{1}{3}[(u+a-x)].\therefore\frac{1}{3}[(u+a-x)]=2x.\therefore x=\frac{1}{7}[u+a]u=a \\
& x=\frac{2}{7}a=\frac{1}{7}[3u-2a]\frac{2}{7}[u+a] \\
& =a=u=eha=300u=400e=10h=20=x=a-x=-u+x-a=-u+x-a-e=\frac{1}{3}[u+x-a-e] \\
& =\frac{2}{3}[u+x-a-e]\frac{1}{3}\times\frac{2}{3}[u+x-a-e]a-x+\frac{1}{3}[u+x-a-e]\frac{1}{3}[a-x]+\frac{2}{3}[u-e]2xx.\therefore x=\frac{3}{25}[7a+2[u-e]-3h]=108
\end{aligned}$$

الخوارزمي موسى بن محمد عبدالله ابو قال

المقابلة الجبر

x شيء

مال

واجب جذر

$$c = \frac{62832}{20000} d\pi = 3.1416$$
